

Solder Mask & Silkscreen

The protective coat + the legend printing — design rules

Two layers that protect, label, and brand your PCB

Audience: Intermediate Engineering Students

LPI mask • Mask colors • Mask openings • Silkscreen rules • Color choices

What are these layers?

Both sit on top of finished copper



Two distinct production layers

Solder Mask

- ▶ Insulating coating over copper
- ▶ Defines pad openings precisely
- ▶ Prevents solder bridges during assembly
- ▶ Protects Cu from oxidation + chemical attack
- ▶ Most common color: green (LPI default)
- ▶ Thickness: 15-30 μm

- ▶ Printed legend on top of mask
- ▶ Shows reference designators, polarity marks, logos
- ▶ Most common color: white on green
- ▶ Ink-based or laser-engraved
- ▶ Defines what's labelled on the board
- ▶ Thickness: 10-20 μm

LPI — Liquid Photo-Imageable Mask

The dominant solder mask technology

- ▶ Wet epoxy ink screen-printed or sprayed across entire board
- ▶ Phototool with pad openings exposed to UV — exposed mask hardens
- ▶ Develop in alkaline solution — unexposed mask in pad areas washes away
- ▶ Final cure at ~150 °C for 60 minutes — durable, non-conductive coating
- ▶ Process used for both single- and double-side mask
- ▶ Resolution: openings down to ~75 µm; gaps between pads ~100 µm
- ▶ Thermal-cured LPI shrinks slightly — fab compensates phototool dimensions

Solder Mask Colors

Green is cheapest — others are cosmetic upcharge

Green

Default, cheapest.
Best silkscreen contrast.
Most forgiving to underlying defects.

Black

Premium look.
White silk most legible.
Underlying traces invisible.

Red

Distinctive (Olimex, Adafruit).
Good contrast w/ white silk.
+₹ over green.

Blue

Industrial / military look.
Good for shielded products.
+₹ over green.

Bright, clean for LED panels.
Hardest to inspect — defects show.
+₹ + slower cure.

Yellow / Purple

Specialty.
Long lead time (some fabs only).
+₹+ premium.

Mask Opening Design Rules

Solder mask defined (SMD) vs Non-solder mask defined (NSMD)

SMD vs NSMD pads

Solder Mask Defined (SMD)

- ▶ Mask opening is SMALLER than the copper pad
- ▶ Mask edge defines the solder area
- ▶ Used for BGA balls (consistent pad size)
- ▶ Better mechanical anchoring of mask
- ▶ Risk: mask overlap reduces effective pad size

Non-Solder Mask Defined (NSMD)

- ▶ Mask opening is LARGER than the copper pad
- ▶ Copper edge defines the solder area
- ▶ Used for standard SMD passives
- ▶ Solder wraps around pad sides — stronger joint
- ▶ Standard for SOIC / TQFP / QFN

Mask Clearance Specs

Manufacturing tolerances + design recommendations

Spec	Standard fab	Premium fab	Notes
Mask opening (NSMD)	Pad + 75 µm/side	Pad + 50 µm/side	Wider = safer registration
Mask opening (SMD)	Pad - 50 µm/side	Pad - 25 µm/side	Tighter = better anchoring
Mask dam between pads	≥ 100 µm	≥ 75 µm	Thinner = more bridging risk
Mask thickness	20-30 µm	15-25 µm	Thicker = better insulation
Mask to trace clearance	100 µm	50 µm	Affects layout density
Mask color uniformity	Acceptable	Best	Premium = consistent across panel

Design tip

Most EDA tools auto-generate solder mask openings = pad + 4 mil (100 µm) per side. Override when needed for fine pitch (drop to 2 mil/50 µm) or for SMD-defined BGA pads.

Silkscreen Design Rules

Legibility comes from following the rules

- ▶ Minimum text height: 0.8 mm (32 mil) — readable in production
- ▶ Minimum line width: 0.15 mm (6 mil) — thinner gets washed out
- ▶ Avoid silk on pads — DRC violation, gets covered by solder
- ▶ Use sans-serif fonts — Calibri / Arial render cleanest
- ▶ Reference designators: visible after assembly
- ▶ Polarity marks: '+' on caps, anode bar on diodes, pin 1 on ICs
- ▶ Board version + date in title block
- ▶ Company logo in silkscreen layer (vector or bitmap)
- ▶ Test point labels (TP1, TP_GND, etc.)
- ▶ Connector pinout (e.g., 'PROG', 'JTAG', 'UART')
- ▶ Mounting hole annotations
- ▶ Compliance marks (CE, FCC, BIS — required for cert)

Silkscreen Printing Methods

Screen print vs inkjet vs laser engraving

Screen print

Traditional method.
Mesh + ink + squeegee.
Cheapest, line width $\geq 100\text{ }\mu\text{m}$.

LPI silkscreen

Liquid photo-imageable.
Same process as solder mask.
Better resolution.

Inkjet (DLI)

Direct legend imaging.
Any pattern, multiple colors.
No phototool needed.

Laser engraving

No ink — burns into mask.
Ultra-fine + permanent.
More expensive.

Two-color silk

White + black on same board.
For branding emphasis.
Not all fabs offer.

Color silk

Yellow, red, blue.
For highlight regions.
Premium upcharge.

Pin-1 Markers & Polarity Indicators

Critical for correct assembly — silkscreen's most important job

- ▶ ICs (SOIC/QFP/QFN): dot, chamfer, or filled triangle near pin 1
- ▶ Polarised caps: '+' symbol near positive terminal
- ▶ Diodes / LEDs: bar/stripe across anode end (or 'A' / 'K' letters)
- ▶ Crystals: pin 1 dot or asymmetric pad geometry
- ▶ Connectors: numbered or labelled (e.g., 'GND, VCC, TX, RX')
- ▶ Use STANDARD symbols — operators recognise them at a glance
- ▶ Common DRC: 'silkscreen-on-pad' — move silk away from pad if it triggers
- ▶ Bonus: print component value next to ref-des (e.g., 'R1 10k', 'C5 100n')

Cosmetic vs Functional Silkscreen

Two different mindsets



Functional silkscreen

- ▶ Ref-des on every component
- ▶ Pin-1 markers on every IC
- ▶ Polarity marks on every polar part
- ▶ Connector pinouts spelled out
- ▶ Mounting hole annotations
- ▶ Test point labels

Cosmetic silkscreen

- ▶ Company logo + board name
- ▶ Board revision (Rev A / B / C)
- ▶ Date or batch code
- ▶ Compliance marks (CE, FCC)
- ▶ QR code (for serial tracking)
- ▶ Engineer's initials (signature)

Common Mistakes

What review catches in 5 minutes

- ▶ Silkscreen over pads → covered by solder, useless after assembly
- ▶ Mask opening too small → poor solderability, hot-air rework fails
- ▶ Mask dam too thin between pads → bridges form during reflow
- ▶ Ref-des under component body → invisible after assembly
- ▶ Mask color = silk color → no contrast, can't read the silk
- ▶ Text height < 0.8 mm → washed out in production, illegible
- ▶ Forgetting to enable silkscreen on BOTH sides of a double-sided PCB
- ▶ Custom logo as bitmap at 100 dpi → blurry; use vector or 1200+ dpi raster

Best Practices

Habits of a clean silkscreen designer

- ▶ Default mask: green (cheapest)
- ▶ Default silk: white (best contrast)
- ▶ Text: ≥ 0.8 mm, sans-serif, vertical or horizontal only
- ▶ Ref-des: above or left of component
- ▶ Pin-1 marker: silkscreen dot AND chamfered footprint corner
- ▶ Title block: company / project / rev / date
- ▶ Visual reference: render at 1:1 in EDA tool before exporting
- ▶ Add: 'top side', 'bottom side' labels for thru-hole-only boards
- ▶ Compliance marks per regulatory body requirements
- ▶ Add mfg / assembly date code in silkscreen for field tracking
- ▶ Use solder-mask 'spy' opening to reveal copper logos under the mask
- ▶ Print test board first — silk often surprises in real ink

Cost Impact

What costs extra in mask + silk

Option	Cost (vs green/white default)
Green mask, white silk	Baseline
Black mask, white silk	+10-30%
Red / Blue / Yellow mask	+30-50%
White mask	+40-60% (cure time, harder to inspect)
Purple / Pink / Specialty	+50-100%, longer lead time
Black silkscreen (on white mask)	Same as white
Multi-color silkscreen	+50-100%
Direct legend imaging (DLI inkjet)	+10-30%
Laser-engraved silk	+50-100% — premium

Key Takeaways

Key takeaways from this lecture

LPI = standard

All modern PCBs use LPI mask.

Green + white = cheapest

Don't pay extra without reason.

Mask opens precisely

NSMD for SMD, SMD for BGA.

Silk ≥ 0.8 mm text

Or it's unreadable.

Functional > cosmetic

Pin-1 + polarity matter most.

Never silk on pads

DRC error + assembly fail.

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- ▶ IPC-SM-840 — Solder mask specification
- ▶ JLCPCB / PCBWay color galleries
- ▶ Phil's Lab — silkscreen design tutorials
- ▶ EDA tool documentation — KiCad / Altium DRC rules
- ▶ Direct Legend Imaging (DLI) — vendor app notes
- ▶ Local fab capability — confirm color availability
- ▶ IPC-A-600 — silkscreen acceptability standards
- ▶ Henkel solder mask formulations — chemistry reference

Thank you • Build something this week.